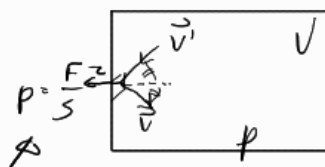


idealni plin $|W_{pot}| < W_{kin}$

vezan sistem $W_{pot} + W_{kin} < 0$

$$W_{pot} \propto \frac{1}{R^6}$$

< < <



tlak je konst. po
celi posodi

$$\Phi_N = \frac{dN}{dt} = h \cdot v_z$$

številka
gostota
delcev,

$$h = \frac{dN}{dv}$$

$$F dt = 2 m_1 v_z dN$$

$\frac{dN}{dt} \rightarrow$ številski tok molekul

$$F = 2 m_1 v_z^2 \cdot \frac{n}{2} \cdot S$$

molekule, ki grejo stran od stene
ne vplivajo na tlak

$$\overline{V^2} = \overline{V_x^2} + \overline{V_y^2} + \overline{V_z^2}$$

- neodvisen
- izotropen
 $\Downarrow$

$$\overline{V^2} = 3 \overline{V_z^2}$$

$$\overline{W_{k1}} = \frac{m_1 \overline{V^2}}{2} = \frac{3}{2} m_1 \overline{V_z^2}$$

$$p = \frac{F}{S} = \frac{2}{3} \frac{N}{V} \overline{W_{k1}}$$

$$\overline{W_k} = \frac{\int dN \cdot W_k}{N = \int dN}$$

$$\frac{N m_1}{M} = \frac{N_A (m)}{M (u)} = \frac{\rho N_A}{M}$$

termodinamsko ravnovesje - Termodin. spr. $(T, spr.) = konst \neq f(t)$
 niso odvisne od časa

ničti

Oti zanka TD:

- če je A v ravnovesju z B in
 B v ravnovesju s C, je tudi
 A v ravnovesju s C.
 (ravnovesje je tranzitivno)

$$\downarrow$$

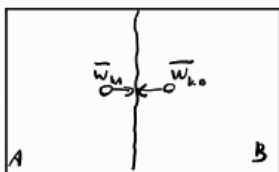
$$T_A = T_B \wedge T_B = T_C \Rightarrow T_A = T_C$$



toplotni stik

- omogoča trke
 molekul iz A in B

- ne omogoča prehoda
 molekul med sistemama



$$T \propto \overline{w_k}$$

makro mikro

$\rightarrow$

$$\overline{w_k} = \frac{3}{2} k_B T \leftarrow \text{(termodinamska definicija temperature)}$$

k_B - Boltzmannova konstanta $[\frac{J}{K}]$

$$k_B = 1,38 \cdot 10^{-23} \frac{J}{K}$$

$$pV = \frac{2}{3} N \cdot \frac{3}{2} k_B T$$

$$pV = N k_B T$$

splošna plinska enačba (za idealen plin)

$$p = n k_B T$$

n
številčna
gostota

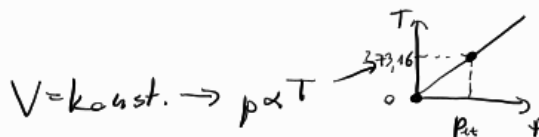
MIKROSKOPSKI
ZAPIS

$$pV = \nu R T$$

$$\nu = \frac{m}{M} = \frac{N}{N_A}$$

$$R = N_A k_B$$

$$pV = \frac{m}{M} R T$$



$$V = konst. \rightarrow p \propto T$$

$$p = konst. \rightarrow V \propto T$$

Trojna točka vode

$$T = 273,16 K$$

$$p = 610 Pa$$

$$0^\circ C = 273,15 K$$

temp. raztezanje

$$\frac{dl}{l} = \alpha dT$$

↑
dolžinskega

linearni koeficient temp. razt. $[K^{-1}]$

- za kovine $\approx 10^{-5} K^{-1}$

$$\frac{dV}{V} = d \ln V = d \ln l^3 = 3 d \ln l = 3 \frac{dl}{l}$$

$$\beta = 3\alpha$$

plini:
(večina so dost idealni)

$$V = \frac{n}{M} R \frac{T}{p}$$

$$\frac{dV}{V} = d \ln V = d(\underbrace{\ln n - \ln M + \ln R + \ln T - \ln p}_0) = \frac{dT}{T} - \frac{dp}{p}$$

↓
 $\beta = \frac{1}{T}$
p = konst.

$$= \beta dT = \chi dp$$

$$\chi = \frac{1}{p}$$

stojivost pri $T = \text{konst.}$

$$\chi = \frac{1}{p} = \frac{\chi p R T}{p M}$$

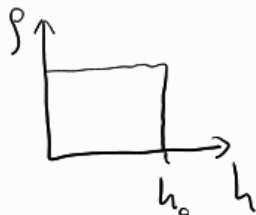
hitrost zvoka v zraku ni odh.
od p , to še ni končna
enačba ker $T \neq \text{konst.}$

Atmosfera

$$p_0 = 10^5 \frac{N}{m^2}$$

za tlak:
mm Hg = torr
milimetri živega srebra

a) $p = \text{konst.}$ model:



$$p_0 = \rho g h_0$$

$$h_0 = \frac{p_0 R T}{\rho g M} = \frac{p_0 R T}{\rho g M}$$

$$= \frac{R T}{M g} \approx 8,5 \text{ km} - \text{ne velja!}$$

(kotaly Mt. Everest)

b) $T = \text{konst.}$

→ atmosfera je v ravnovesju

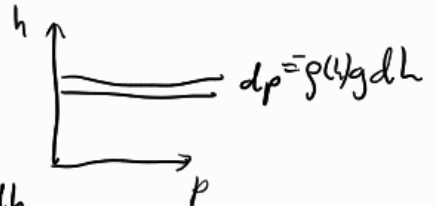
$$\text{realno: } \frac{\Delta T}{T} \approx 10\%$$

$$\rho(h) \propto p(h)$$

$$dp = -\frac{\rho M}{R T} g dh$$

$$\frac{dp}{p} = -\frac{M g}{R T} dh = -\frac{dh}{h_0}$$

$$p = p_0 e^{-\frac{h}{h_0}} \text{ barometrska enačba}$$



$$p = p_0 e^{-\frac{h}{h_0}} = p_0 e^{-\frac{N}{N_0} \frac{Mg}{RT} h} = p_0 e^{-\frac{m_p g h}{k_B T}}$$

molekule so raje spodaj kot zgoraj zaradi W_p

↓
termična energija

$$u(h) = u_0 e^{-\frac{W_p}{k_B T}} = \frac{dN}{dV} = \frac{dN}{S dh}$$

$$\frac{dN}{dV} \propto e^{-\frac{W_p}{k_B T}}$$

podobna lahko rečemo tudi za W_k

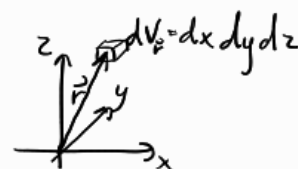
Boltzmanova porazdelitev

$$f(W_k) \propto e^{-\frac{W_k}{k_B T}} = e^{-\frac{m_1 v^2}{2 k_B T}}$$

velja v sistemih, ki so v termodin. ravnovesju

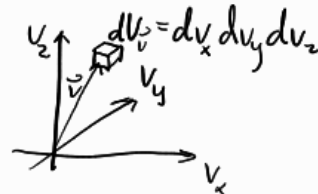


$W_p \rightarrow$ lega $\vec{r} \rightarrow$ homogen prostor $\vec{r}$

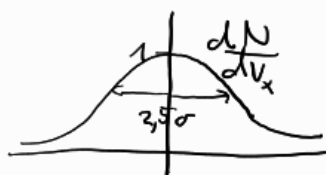


$W_k \rightarrow$ hitrost $\vec{v} \rightarrow$ homogen prostor $\vec{v}$

$$\frac{dN}{dv_x} = \text{konst.} \cdot f(W) = \text{konst.} \cdot e^{-\frac{m_1 v_x^2}{2 k_B T}}$$



$$\sigma = \sqrt{\frac{k_B T}{m_1}}$$



$$\frac{dV}{dv} = \frac{dV}{dv_x dv_y dv_z} \cdot \frac{dv_x dv_y dv_z}{dv} = \text{konst.} \cdot 4\pi v^2 \cdot \frac{4\pi v^2 dv}{dv}$$

$$\frac{dv^2 e^{-\frac{v^2}{2\sigma^2}}}{dv} = 0$$

$$= e^{-\frac{v^2}{2\sigma^2}} \left(2v - v^2 \frac{2v}{2\sigma^2} \right) \Rightarrow$$

$$\Rightarrow v_{mp} = \sqrt{2} \sigma \quad \sigma = \sqrt{\frac{2 k_B T}{m_1}}$$

most probable value

$$\langle v^2 \rangle = \frac{3 k_B T}{m_1} = 3 \sigma^2$$

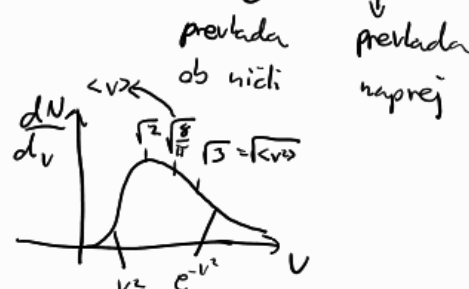
$$\sqrt{\langle v^2 \rangle} = \sqrt{3} \sigma$$

NORMALIZACIJSKA KONSTANTA

$$\frac{dN}{dv} = \frac{dN}{d^3 \vec{v}} \cdot \frac{d^3 \vec{v}}{dv} \cdot f(W_k) = A \cdot 4\pi v^2 e^{-\frac{m_1 v^2}{2 k_B T}}$$

konst. $4\pi v^2$

$$A = \left(\frac{m_1}{2\pi k_B T} \right)^{3/2}$$



Maxwell-Boltzmanova porazdelitev

$$j_v = \frac{dN}{dt} \frac{1}{S} = n \langle v_z \rangle = 0$$

isto jih gre v
eno smer kot v drugo



$$j_v = \frac{1}{2} n \langle v_z \rangle$$

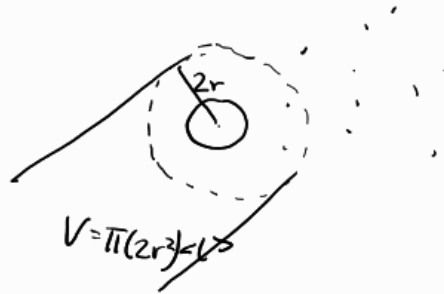
ocena:

$$\frac{n}{6} \langle v \rangle < j_v < \frac{n}{2} \langle v \rangle$$

realno

$$j_v = \frac{n}{4} \langle v \rangle$$

Povprečna prosta pot $\langle l \rangle$



trk se v povprečju zgodi, ko:

$$nV = 1$$

$$n \pi (2r)^2 \langle l \rangle = 1$$

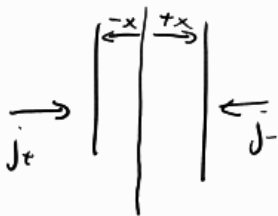
$\langle l \rangle = \frac{1}{n \pi (2r)^2}$ - velja, če bi se ena molekula gibala, ostale pa mirovale

$$\langle l \rangle = \frac{1}{\sqrt{2} n \pi (2r)^2}$$

$$n = \frac{p}{k_B T} = \frac{10^5 \text{ N/m}^2}{1,4 \cdot 10^{-23} \text{ J/K} \cdot 300 \text{ K}} = 2 \cdot 10^{25} \text{ m}^{-3}$$

$$\langle l \rangle = \frac{1 \text{ m}}{2 \cdot 10^{25} \cdot 10^{-19} \text{ m}^2 \cdot 1,4 \cdot 10} = 10^{-7} \text{ m}$$

Difuzijski tok



$$j_x = j_+ - j_- = \frac{\langle v \rangle}{4} (n_+ - n_-) = \frac{\langle v \rangle}{4} 2x \frac{dn}{dx}$$

tak bo neničeln, če je
na obeh straneh drugačna
koncentracija molekul

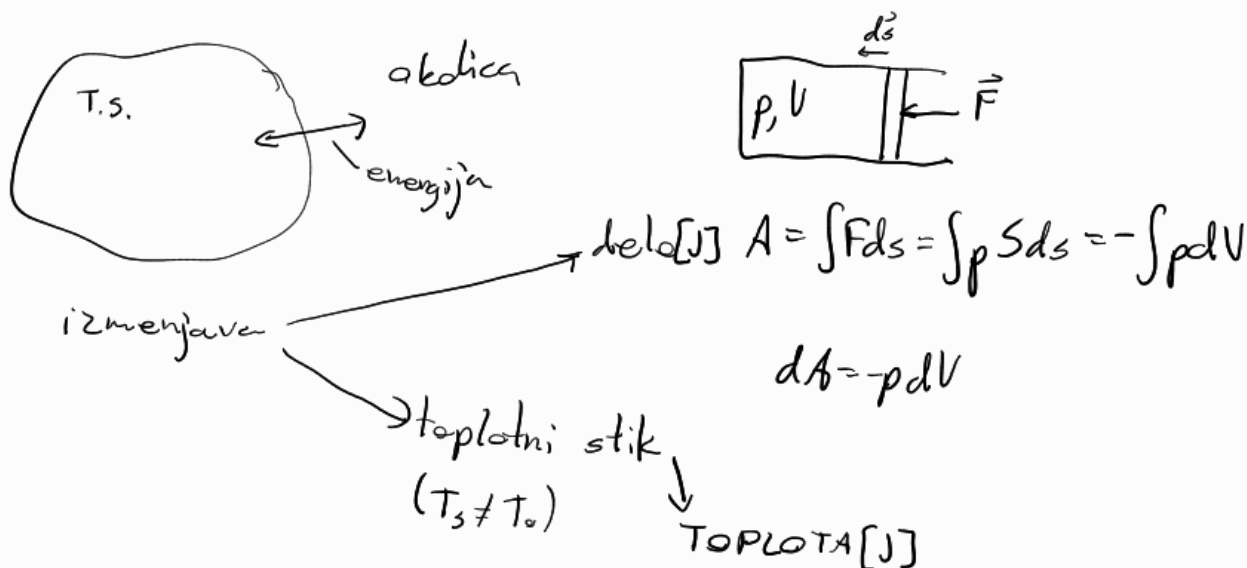
$$n_+ - n_- = \frac{dn}{dx} 2x$$

$$j_{dt} = -D \frac{dn}{dx}$$

difuzijski koeficient

$$D = \frac{\langle l \rangle \langle v \rangle}{3}$$

Difuzija je zelo počasna



1. ZAKON TERMODIN. SIST.

$$\Delta W_n = A + Q$$

Idealni plin

enoatomni plin

$$\Delta W_n = N \langle W_k \rangle = N \frac{3}{2} k_B \Delta T$$

$$\Delta W_n = m c_v \Delta T = Q$$

pri $V = \text{konst}$ ($A = - \int p dV = 0$)

enoatomni plin:

$$dW_n = m c_v dT = N \frac{3}{2} k_B dT$$

$$c_v = \frac{N}{m} \frac{3}{2} k_B = \frac{3}{2} \frac{N_A \cdot k_B}{M} = \boxed{\frac{3}{2} \frac{R}{M}}$$

$$\boxed{C_M = c_v M = \frac{3}{2} R} = \text{konst. za vse enoatomne pline}$$

↑
MOLARNA SPEC. TOPL.

dvoatomne molekule

$$\frac{3}{2} \rightarrow \frac{5}{2}$$



vrtenje
okrog simetrijskih osi
ne prinese dodatne
 $\frac{1}{2}$

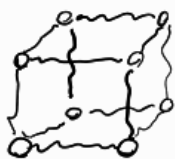
$$M c_v = \frac{5}{2} R$$

voda (para):

$$M c_v = 3 R$$

$$CO_2: \frac{5}{2}$$

Kristali (trdne snovi)



$M_{c(v)} = 3R$
 ↑
 ni treba, ker
 se V pri segrevanju
 ne spremeni veliko
 $\frac{3}{2}$ od translacijske kinetične
 $\frac{3}{2}$ od potencialne
 $\approx 25 \text{ kJ/K}$

	c	M
Pb	130	207
Hg	138	201
Cu	390	64
Al	880	27

$$m \cancel{c_p} \Delta T = Q_{p=k} = \Delta W_n - A = m \cancel{c_v} \Delta T + m \frac{R}{M} \Delta T$$

$$A = - \int p dV = - p(V - V') = - \frac{m}{M} R \Delta T$$

konst.

$$\Rightarrow \boxed{c_p - c_v = \frac{R}{M}} \text{ ZA IDEALNI PLIN}$$

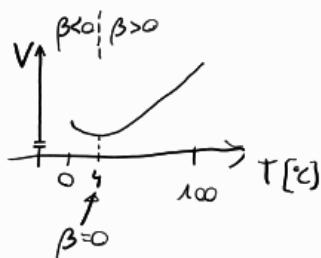
$$\boxed{\chi = \frac{c_p}{c_v}} = \frac{N_{ps} + 2}{N_{ps}} \rightarrow \frac{1at}{2at} \dots \frac{5}{6}$$

3t. prostih
stopinj

za kovine
 $c_p \approx c_v = c$

VODA

$$\frac{dV}{V} = \beta dT$$



$$c = 4200 \frac{\text{J}}{\text{kg K}} \quad \frac{R}{M} \approx 500$$

$$\Rightarrow \boxed{N_{ps} = 17 ??}$$

zaradi vodikovih vezi

Entalpija

$$\Delta H = \Delta W_n + p \Delta V$$

ni pravi diferencial
↓

$$dH = dW_n + d(pV) = dQ - p dV + dpV + p dV$$

$$dH = dQ + V dp \rightarrow dH = (dQ)_{p=k}$$

Idealni plin

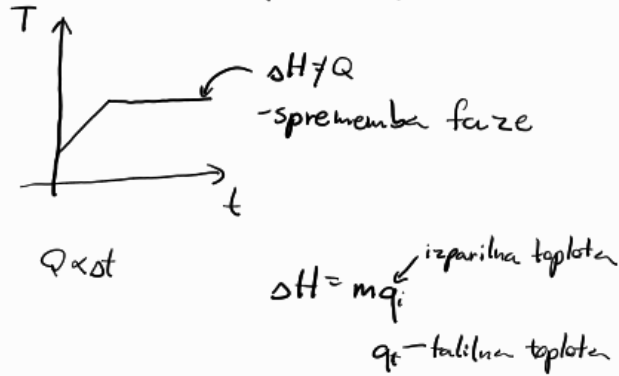
$\Delta H = m c_p \Delta T \rightarrow$ problem enačbe pri kapljevinah/trdninah:

~za vodo

$$c = 4200 \frac{\text{J}}{\text{kgK}}$$

$$q_f = 236 \frac{\text{kJ}}{\text{kg}}$$

$$q_i = 224 \frac{\text{MJ}}{\text{kg}}$$



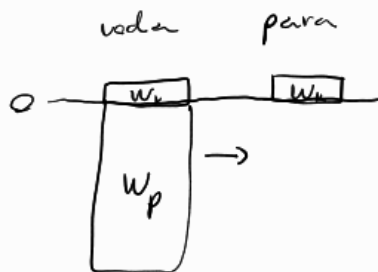
Voda vre:

$$\Delta W_n = Q + A = \Delta H - p(V - V')$$

$\uparrow$ $\uparrow$
 1700 l 1 l
 1 kg vode

za 1 kg vode

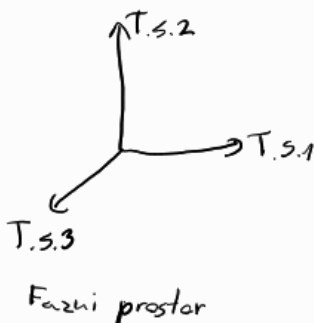
$$\Delta W_n = 2,24 \text{ MJ} - 0,17 \text{ MJ}$$



Metastabilno stanje - tekoča voda pod lediščem - podhlajena voda
(čisto voda namo na miru in počasi hladimo)

- tekoča voda nad vreliščem - pregreta voda
- diamant pri $T = 20^\circ\text{C}$ (zelo počasi se spreminja v grafit)

stanje t. sist. $\rightarrow$ nabor T. spr.



Idealni plin: $p, V, T \rightarrow$ 3D fazni prostor

skupje prostora

plinska enačba $\rightarrow$ enačba stanja

prostor $\rightarrow$ ploskev

3D $\rightarrow$ 2D

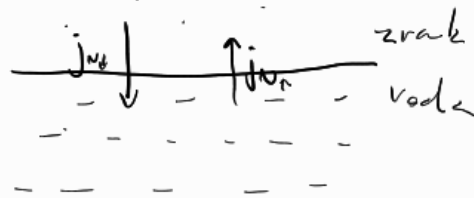
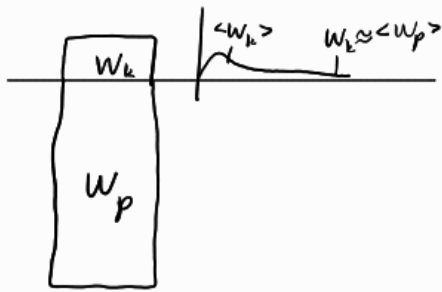
Mešanica

$\hookrightarrow$ idealnih plinov

- delni tlaki p_i

$$p = \sum p_i$$

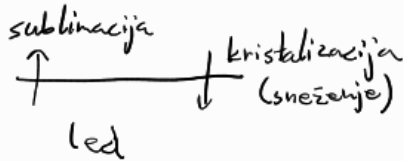
$$p_i V = \frac{m_i}{M_i} R T$$



$$j_{vd} = n \frac{v^2}{4}$$

$$p_{H_2O} = \frac{m_{H_2O}}{M_{H_2O}} \frac{1}{V} RT$$

meja - podana z nasičenim parnim tlakom ($j_{vd} = j_{vz}$)



Izohorna sprememba	Izobarna sprememba	Izotermna sprememba
$V = \text{konst.}$ $\frac{P}{T} = \text{konst.}$ $A = 0$ $\Delta W_h = Q = m c_v \Delta T$	$p = \text{konst.}$ $\frac{V}{T} = \text{konst.}$ $A = -p(V - V')$ $Q = m c_p \Delta T$	$T = \text{konst.}$ $pV = \text{konst.}$ $\Delta W_h = 0$ $Q = -A$ $A = -\int p dV = -\int_{V'}^V \frac{p_0 V_0}{V} dV =$ $= -\frac{m}{M} RT \ln \frac{V}{V'} \Rightarrow$ $Q = \frac{m}{M} RT \ln \frac{V}{V'}$

$$\gamma = \frac{c_p}{c_v}$$

$$c_p - c_v = \frac{R}{M}$$

Adiabatsna sprememba

$$Q = 0$$

$$pV = \frac{m}{M} RT$$

$$dQ = 0 = dW_h - dA = m c_v dT + p dV$$

$$m c_v dT = -p dV = -\frac{m}{M} \frac{dV}{V} RT$$

$$\frac{dT}{T} = -\frac{R}{M c_v} \frac{dV}{V}$$

$$\frac{c_p - c_v}{c_v} = \gamma - 1$$

$$pV^{\gamma-1} = [pV^{\gamma}] = \text{konst.}$$

$$\ln \frac{T}{T_0} = -(\gamma - 1) \ln \frac{V}{V_0} = \ln \left(\frac{V_0}{V} \right)^{\gamma-1}$$

$$T \left(\frac{T}{p} \right)^{\gamma-1} = \left[\frac{T^{\gamma}}{p^{\gamma-1}} \right] = \text{konst.}$$

$$TV^{\gamma-1} = T_0 V_0^{\gamma-1}$$

$$dW_n = dQ + dA$$

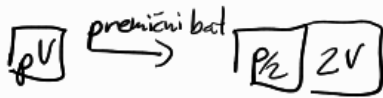
$$dA = -pdV$$

$$dQ = Tds$$

$$dS - \text{ENTROPJA} \left[\frac{J}{K} \right]$$

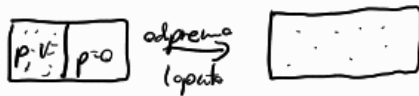
$$\boxed{dS \geq \frac{dQ}{T}} \quad \text{2. ZAKON T.D.}$$

Adiabatsna spr. $\rightarrow$ IZENTROPNA SPREMEMBA



$$\Delta S = \int dS = \int \frac{dQ}{T} = \int \frac{m R T dV}{m T V} = \frac{m}{M} R \ln \frac{V}{V_0}$$

obrnljiva
sprememba
(reverzibilna)



$$\Delta S = 0 ???$$

neobrnljiva spr.
(ireverzibilna)

$dS = \frac{dQ}{T}$ $dA = -pdV$ velja za obrnljivo spr.	$dS > \frac{dQ}{T}$ $dA > -pdV$ velja za neobrnljivo spr.
--	--

Entropija?

$$S = k_B \ln W$$

neka ogramna številka?



ni možno?

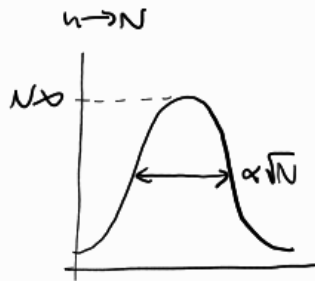
- verjetnost, da se vse molekule nahajajo na levi strani posode? v nekem trenutku

- 6 metrov kovanca:

makro	0-6	1	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$
	1-5	6	
	2-4	15	
	3-3	20	
	4-2	15	
	5-1	6	
	6-0	1	

$$64 = 2^6 = W$$

↓ mikro

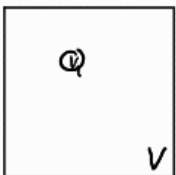


↓
1
2^{št. molekul}
↓
praktično ni možno.

$$S = k_B \ln W; W \approx 2^N$$

$$\begin{matrix} N \rightarrow 2N \\ W \rightarrow W^2 \\ \downarrow \ln \quad \downarrow \\ S \rightarrow 2S \end{matrix}$$

Dajemo po 1 molekulo v posodo



$$\begin{matrix} 1 \\ \vdots \\ 2 \\ \vdots \\ N \end{matrix} \left| \begin{matrix} \frac{V}{V_1} \\ \frac{V-V_1}{V_1} \\ \vdots \\ \frac{V-NV_1}{V_1} \end{matrix} \right.$$

$$\rightarrow W = \prod_{i=1}^N w_i \stackrel{\text{ker je idealni plin}}{=} w_1^N = \left(\frac{V}{V_1}\right)^N$$



$$S = k_B \ln W = N k_B \ln \frac{V}{V_1}$$

$$\Delta S = N k_B \ln \frac{V}{V_1} = \frac{m}{M} R \ln \frac{V}{V_1}$$

↑
pri izotermni spr.

